DESSERTOPIA

Order Management And Order Details System

INTRODUCTION :

- The primary objective of this report is to design, implement, and document a robust relational database schema for an Order Management and Order Details system using SQL.
- The system is structured to capture customer purchasing behavior, maintain granular line-item records of products ordered, ensure strict data integrity, and support efficient administrative data retrieval.

Creation of the ORDERS table:

```
SQL> CREATE TABLE ORDERS (
  2      ORDER_ID VARCHAR2(10) PRIMARY KEY,
  3      CUSTOMER_ID VARCHAR2(10),
  4      ORDER_DATE DATE,
  5      TOTAL_AMOUNT NUMBER(10,2),
  6      ORDER_STATUS VARCHAR2(20),
  7      FOREIGN KEY (CUSTOMER_ID)
  8          REFERENCES CUSTOMER(CUSTOMER_ID)
  9  );

Table created.
```

Data Insertion: ORDER VALUES

```
SQL> INSERT INTO ORDERS VALUES
  2  ('O001', 'C001', DATE '2026-10-01', 550, 'Pending');

1 row created.
```

```
SQL> INSERT INTO ORDERS VALUES
  2  ('O001', 'C001', DATE '2026-10-01', 550, 'Pending');

1 row created.
```

Creation of the ORDER_DETAILS table :

```
SQL> CREATE TABLE ORDER_DETAILS (
  2      ORDER_DETAIL_ID VARCHAR2(10) PRIMARY KEY,
  3      ORDER_ID VARCHAR2(10),
  4      PRODUCT_ID VARCHAR2(10),
  5      QUANTITY NUMBER,
  6      PRICE NUMBER(10,2),
  7      FOREIGN KEY (ORDER_ID) REFERENCES ORDERS(ORDER_ID),
  8      FOREIGN KEY (PRODUCT_ID) REFERENCES PRODUCT(PRODUCT_ID)
  9  );

Table created.
```

Data Insertion : ORDER_DETAIL VALUES

```
SQL> INSERT INTO ORDER_DETAILS VALUES
  2  ('OD001', 'O001', 'P001', 1, 550);

1 row created.
```

```
SQL> INSERT INTO ORDER_DETAILS VALUES
  2  ('OD002', 'O002', 'P002', 2, 120);

1 row created.
```

Table structures using DESC :

```
SQL> DESC ORDERS;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ORDER_ID                                  NOT NULL VARCHAR2(10)
 CUSTOMER_ID                                        VARCHAR2(10)
 ORDER_DATE                                         DATE
 TOTAL_AMOUNT                                       NUMBER(10,2)
 ORDER_STATUS                                       VARCHAR2(20)
```

```
SQL> DESC ORDER_DETAILS;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ORDER_DETAIL_ID                           NOT NULL VARCHAR2(10)
 ORDER_ID                                           VARCHAR2(10)
 PRODUCT_ID                                         VARCHAR2(10)
 QUANTITY                                           NUMBER
 PRICE                                              NUMBER(10,2)
```

Resulting Data :

ORDERS now holds seven rows:

```
SQL> SELECT* FROM ORDERS ORDER BY ORDER_ID;

ORDER_ID   CUSTOMER_I ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ---------- --------- ------------ --------------------
O001       C001       01-OCT-26          550 Pending
O002       C002       02-OCT-26          240 Shipped
O003       C003       03-OCT-26          360 Delivered
O004       C004       04-OCT-26          300 Pending
O005       C005       05-OCT-26          500 Shipped
O006       C006       06-OCT-26          450 Delivered
O007       C007       07-OCT-26          400 Pending

7 rows selected.
```

ORDER_DETAILS also holds seven rows, one line item per order:

```
SQL> SELECT* FROM ORDER_DETAILS ORDER BY ORDER_DETAIL_ID;

ORDER_DETA ORDER_ID   PRODUCT_ID   QUANTITY      PRICE
---------- ---------- ---------- ---------- ----------
OD001      O001       P001                1        550
OD002      O002       P002                2        120
OD003      O003       P003                2        180
OD004      O004       P004                2        150
OD005      O005       P005                5        100
OD006      O006       P006                1        450
OD007      O007       P007                2        200

7 rows selected.
```

INTEGRITY CONSTRAINTS :

```
SQL> ALTER TABLE ORDERS
  2  MODIFY CUSTOMER_ID NOT NULL;

Table altered.
```

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Table altered.
```

```
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  2  MODIFY CUSTOMER_ID NOT NULL;

Table altered.
```

CHECK CONSTRAINTS :

```
SQL> ALTER TABLE ORDERS
  2  ADD CONSTRAINT CK_ORDER_AMOUNT
  3  CHECK (TOTAL_AMOUNT >= 0);

Table altered.
```

```
SQL> ALTER TABLE ORDERS
  2  ADD CONSTRAINT CK_ORDER_STATUS
  3  CHECK (ORDER_STATUS IN ('Pending', 'Shipped', 'Delivered'))
;

Table altered.
```

```
SQL> ALTER TABLE ORDER_DETAILS
  2  ADD CONSTRAINT CK_ORDER_QUANTITY
  3  CHECK (QUANTITY > 0);

Table altered.
```

**** THANK YOU ****